

# **MUST lecture 6**

## **Univariate Multiple Linear Regression**

### **(UMLR)**

#### **Disclaimer:**

**Please note that these slides are only a brief overview of the topic.  
Many more details will be explained on the blackboard and with  
MATLAB examples**

# UMLR, Univariate Multiple Linear Regression

**Model**  $Y = \beta_0 + \beta_1 z_1 + \cdots + \beta_r z_r + \varepsilon$

$Y$  = univariate response variable (known/measured)

$z_1, \dots, z_r$  = predictor variables with known (set) values

$\beta_0, \beta_1, \dots, \beta_r$  = unknown regression coefficients (marginal slopes)

$\varepsilon$  = random error term for non-perfect model (unknown)

**Assumptions**  $E[\varepsilon] = 0, \quad V[\varepsilon] = \sigma^2$  (constant but unknown),  $Cov[\varepsilon_i, \varepsilon_j] = 0$

**Note:** no assumption on distributions

**n observations** 
$$\begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix} = \begin{bmatrix} 1 & z_{11} & \cdots & z_{1r} \\ 1 & z_{21} & \cdots & z_{2r} \\ \vdots & \vdots & & \vdots \\ 1 & z_{n1} & \cdots & z_{nr} \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_r \end{bmatrix} + \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{bmatrix}$$

$$Y = Z \beta + \varepsilon$$

$(n \times 1) = (n \times (r+1)) ((r+1) \times 1) + (n \times 1)$

## Least Squares estimation of regression coefficients

**Residuals**  $\hat{\varepsilon}_j = Y_j - \hat{Y}_j = Y_j - z_j \hat{\beta}$

**Cost function**  $SSE(\hat{\beta}) = \sum_{j=1}^n \hat{\varepsilon}_j^2 = \hat{\varepsilon}^T \hat{\varepsilon} = (Y_j - z_j \hat{\beta})^T (Y_j - z_j \hat{\beta})$

Minimizing:  $\vec{\nabla} SSE(\hat{\beta}) = 0$

giving  $\hat{\beta} = (Z^T Z)^{-1} Z^T Y$  (LS estimate)

$$E[\hat{\beta}] = \beta \quad (\text{unbiased estimate})$$

$$Cov[\hat{\beta}] = \sigma^2 (Z^T Z)^{-1} \quad (\text{consistent estimate})$$

**Projections**  $\hat{Y} = Z \hat{\beta} = Z (Z^T Z)^{-1} Z^T Y = HY, \quad H = \text{projection matrix}$

**Variance estimate**  $\hat{\sigma}^2 = \frac{SSE}{n-(r+1)} = \frac{1}{n-(r+1)} \sum_{j=1}^n \hat{\varepsilon}_j^2 = \frac{1}{n-(r+1)} \hat{\varepsilon}^T \hat{\varepsilon}$

**SS decomposition**  $SST = SSE + SSR$   
 $SST = \sum_{j=1}^n (Y_j - \bar{Y})^2, \quad SSE = \sum_{j=1}^n (Y_j - \hat{Y}_j)^2, \quad SSR = \sum_{j=1}^n (\hat{Y}_j - \bar{Y})^2$

**Model fit measure**  $R^2 \stackrel{\text{def}}{=} \frac{SSR}{SST} = 1 - \frac{SSE}{SST}, \quad 0 \leq R^2 \leq 1$

## Inference for estimated regression coefficients

New assumption on distribution

$$\varepsilon \sim N_n(\mathbf{0}, \sigma^2 I_{nn})$$

From this follows directly

$$Y \sim N_n(\mathbf{Z}\boldsymbol{\beta}, \sigma^2 I_{nn})$$

$$\hat{\boldsymbol{\beta}} \sim N_{r+1}(\boldsymbol{\beta}, \sigma^2 (\mathbf{Z}^T \mathbf{Z})^{-1})$$

$$\hat{\sigma}^2 \sim \frac{\sigma^2}{n-(r+1)} \chi^2(n - (r + 1))$$

**100(1 -  $\alpha$ )% Confidence region for  $\boldsymbol{\beta}$ :**  $\frac{(\boldsymbol{\beta} - \hat{\boldsymbol{\beta}})^T \mathbf{Z}^T \mathbf{Z} (\boldsymbol{\beta} - \hat{\boldsymbol{\beta}})}{s^2(r+1)} \leq F(r + 1, n - (r + 1))_\alpha$

**100(1 -  $\alpha$ )% Bonferroni CI for  $\beta_i$ :**  $\left[ \hat{\beta}_i \pm s_{\hat{\beta}_i} t(n - (r + 1))_{\frac{\alpha/2}{r+1}} \right] \quad i = 1, \dots, r + 1$   
 $s_{\hat{\beta}_i} = \sqrt{\{\hat{\sigma}^2 (\mathbf{Z}^T \mathbf{Z})^{-1}\}_{ii}}$

## Hypothesis tests for regression coefficients

Null hypothesis  $H_0: \beta = \beta_0$

Test statistic  $\frac{(\hat{\beta} - \beta_0)^T Z^T Z (\hat{\beta} - \beta_0)}{s^2(r+1)} \sim F(r+1, n - (r+1))$

## Extra Sum of Squares (ESS) test for significant predictor variables

Split up  $\beta = \begin{bmatrix} \beta_{(1)} \\ \beta_{(2)} \end{bmatrix}$  where  $\beta$  is  $(r+1) \times 1$ ,  $\beta_{(1)}$  is  $(q+1) \times 1$ ,  $\beta_{(2)}$  is  $(r-q) \times 1$

$H_0: \beta_{(2)} = 0$  (meaning corresponding predictor variables are insignificant)

$$SST = SSE(\beta) + SSR(\beta) = SSE(\beta_{(1)}) + SSR(\beta_{(1)}) \quad (\text{constant sum})$$

$$ESS(\beta_{(1)} \rightarrow \beta) = SSR(\beta) - SSR(\beta_{(1)}) = SSE(\beta_{(1)}) - SSE(\beta) \geq 0$$

Test statistic  $\frac{ESS(\beta_{(1)} \rightarrow \beta) / \Delta df}{\hat{\sigma}^2} = \frac{ESS(\beta_{(1)} \rightarrow \beta) / (r-q)}{\hat{\sigma}^2} \sim F(r-q, n - (r+1))$

Special case:  $q = 0$  corresponding to testing  $H_0: \beta_1 = \dots = \beta_r = 0$   
(no dependency of Y on any predictor variables = model not significant)

## Prediction of response at new value of predictor variables $z = z_0$

Prediction of mean response

$$E[Y(z_0)] = z_0^T \beta$$

estimator

$$z_0^T \hat{\beta} \sim N_1(z_0^T \beta, z_0^T (Z^T Z)^{-1} z_0 \sigma^2)$$

CI for mean response

$$\left[ z_0^T \hat{\beta} \pm \sqrt{z_0^T (Z^T Z)^{-1} z_0 \hat{\sigma}^2} t(n - (r + 1))_{\alpha/2} \right]$$

Prediction of single response

$$Y(z_0) \sim N_1(z_0^T \beta, \sigma^2)$$

$$E[Y(z_0)] \sim N_1(z_0^T \beta, z_0^T (Z^T Z)^{-1} z_0 \sigma^2)$$

$$Y(z_0) - E[Y(z_0)] \sim N_1(0, (1 + z_0^T (Z^T Z)^{-1} z_0) \sigma^2)$$

CI for single response

$$\left[ z_0^T \hat{\beta} \pm \sqrt{(1 + z_0^T (Z^T Z)^{-1} z_0) \hat{\sigma}^2} t(n - (r + 1))_{\alpha/2} \right]$$